Inexpensive* High-Throughput Computer-Aided Tracking of Zebrafish in the Novel Tank Paradigm

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*assuming you have Mathematica

Abstract

Zebrafish (*Danio Rerio*) have become an important model organism in biomedical and neuroscience research. For researchers interested in neural developmental, including genetic and teratological influences, Zebrafish offer several advantages over laboratory rodents, including cost of maintenance, speed of development, and transparent embryos that develop outside the womb that can easily be imaged and experimentally manipulated. Further, since Zebrafish reach adulthood in several months' time, it is also relatively easy to examine the long-term impact of developmental disturbances.

Zebrafish have been used extensively to study teratological influences on anxiety-like behaviors. Anxiety-like behaviors include thigmotaxis (edge-preference) in juvenile fish and bottom preference in adult fish. These behaviors, when exhibited in novel environments, are thought to be analogous to rodent behaviors in open field tests. Pharmacological manipulations with anxiogenics and anxiolytics produce reliable alterations in these behaviors consistent with the interpretation that these behaviors reflect anxiety-like states in the fish. Experimental setups for assessing anxiety-like behaviors typically involve either still photography at regular intervals or continuous videotaping followed by selective coding of behaviors. In many cases behavior is coded discreetly (fish is touching edge, fish is located in top 25% of tank).

The current project sought to develop an automated, inexpensive, general-purpose tracking system that would allow for high-throughput, reliable, and comprehensive assessment of juvenile and adult behaviors. Using relatively inexpensive hardware and modifications to widely-available software, we developed a system that produces high-quality and fine-grained continuous assessments of fish location allowing for both traditional behavioral assessments, as well as outcomes such as average velocity and frequency of changes in direction. To demonstrate the capabilities of the system we present evidence of test-retest reliability for a variety of behavioral outcomes in 103 adult zebrafish in the novel tank paradigm.

Modified Novel Tank

The Novel Tank Paradigm is a widely used behavioral assay of anxiety-related behaviors in Zebrafish. It is similar to open-field tests used with rodents. In the Novel Tank Procedure, Zebrafish are introduced to a novel tank, often following administration of anxiolytics or anxiogenics, and their position within the tank is assessed over some period of time. Major outcomes traditionally assessed include latency to enter the top, time spent in the top, frequency and duration of freezing behavior and erratic swimming. Typically less time spent in the top of the tank, more freezing, and more erratic swimming are interpreted as evidence of increased anxiety.

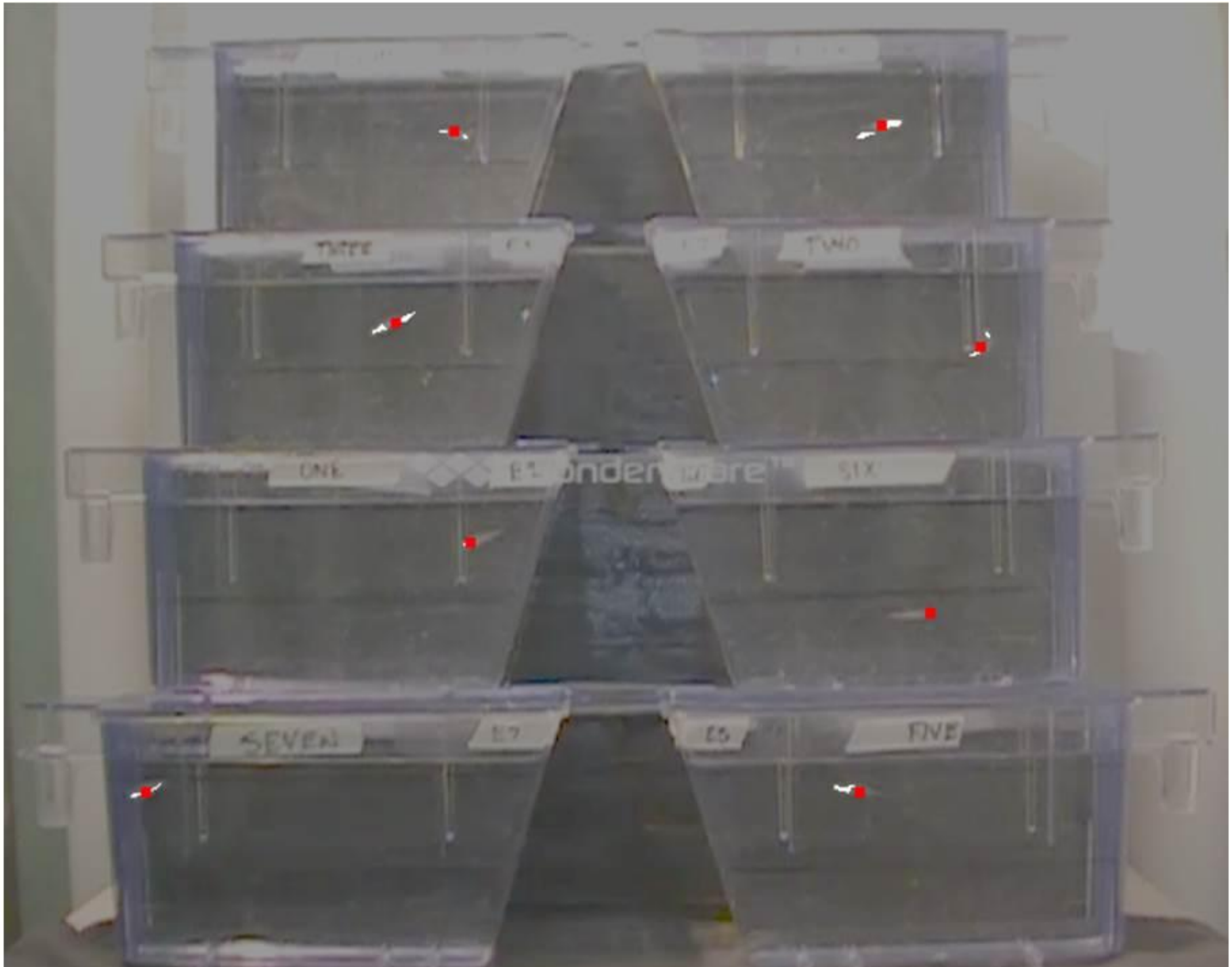
Zebrafish behaviors are typically recorded on video and then coded by humans or computers. Computerized coding greatly reduces the labor involved in coding but widely-available commercial behavior coding software tends to be expensive.

The current project sought to develop software and an experimental setup to allow for inexpensive, automated, high through-put assessments of zebrafish in the Novel Tank Paradigm.

The system we developed allowed us to processes a video of 8 Zebrafish in separate tanks. Each video was processed using custom written Mathematica code. The code produces information about each fish’s x and y coordinate location 6 times per second.

Figure 1

Mathematica was used to identify centroids for each fish for each frame.



Testing the System: Reliability

As a demonstration of the system we chose to examine the test-retest reliability of a few select behaviors in the Novel Tank Paradigm. We tested 103 fish in the Novel Tank Paradigm twice a day (morning and afternoon) for three days producing 618 assessments. Assessments were 20 minutes in duration. From these assessments, our system generated 4.8 million x,y coordinate pairs. These coordinate pairs were then reduced in SPSS to produce the following outcomes: % of time in top of tank, latency to enter top of tank, mean vertical position in tank, and the skew of the vertical position in the tank.

Cronbach's Alphas

Latency to Entry to Top Half:	.77
Mean Vertical Position in Tank:	.71
Percentage of Time in Top Half:	.86
Skew of Vertical Position in Tank:	.85

Interrater Reliability (Human, Computer)

Percentage of Time in Top Half (N=24): r=.91

Conclusion

There were two main goals of the current project. First, we sought to develop an automated, inexpensive, general-purpose tracking system that would allow for high-throughput, reliable, and comprehensive assessment of Zebrafish behaviors in the Novel Tank Paradigm. The high level of agreement (inter-rater reliability) between our human coder and the software system we developed supports the accuracy and utility of our software coding system. Future work on the system should further improve its accuracy and allow us to examine a wider range of behaviors including distance travelled, acceleration, and behavioral freezing.

Second, we sought to establish the test-retest reliability of bottom-dwelling behavior in the Novel Tank paradigm. We found evidence of strong stability across multiple points in time. These results suggest that the Novel Tank Paradigm is sensitive not just to mean-level differences among groups (e.g., treatment vs. control groups), but can also be used to examine individual-level variation in fish behaviors.